

Kishaa International Platform Specification

Comprehensive System Architecture, Autonomous Editorial Publishing Pipelines, Real-Time Government Rate Verification, Cloud Storage CDN Infrastructure, and Multi-Office Operations Suite.

PLATFORM ARCHITECTURE

Node.js v22 · Express SSR · React 19 SPA · tRPC · SQLite
WAL · Supabase Cloud Storage CDN & PostgreSQL

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ORGANIZATION & GOVERNANCE

Kishaa International (Pvt) Ltd.
Chief Immigration Counsel: Zeb Khan
Operating Desks: Islamabad · Lahore · London · Dubai

SECURITY & COMPLIANCE

Multi-Layer Anti-Indexing Defense · Cryptographic Session
Persistence · Trustpilot Automated Invitation Hook

1. Executive Summary & Core Architecture

Kishaa International is a premier international education consultancy and licensed immigration practice founded and directed by Chief Immigration Counsel **Zeb Khan**. Operating physical advisory desks across Islamabad, Lahore, London, and Dubai, the firm specializes in university admissions, statutory proof-of-funds structuring, visit visa documentation, business investor routes, and Cambridge/IELTS testing preparation.

The Kishaa International digital platform represents a modern hybrid architecture that unifies high-performance server-side rendering (SSR) for instant discoverability and Googlebot indexing compliance with an ultra-responsive React 19 single-page application (SPA) client for interactive student tools.

Answer-First Editorial Principle: The platform enforces transparent immigration guidance. Every guide, maintenance requirement, blocked account threshold, and living cost breakdown explicitly states exact figures, effective regulatory dates, and statutory legal citations upfront.

1.1 Technology Stack Architecture

Layer	Technologies Employed	Architectural Purpose
Frontend Client	React 19, TypeScript, Tailwind CSS, Vite 7, Lucide Icons, React Router 7, TanStack Query	Client-side routing, interactive consultation modals, dynamic cost planners, and real-time state management.
SSR & Rendering	Node.js v22, Express.js, Handlebars.js, CSS variables design system	Ultra-fast initial HTML delivery, complete schema.org structured data, and search engine crawling compliance.
API & Data Layer	tRPC (TypeScript-first RPC), Express JSON REST controllers	End-to-end typed contract connecting client forms with backend database queries and validation controllers.
Local Persistence	Embedded SQLite (better-sqlite3) with Write-Ahead Logging (WAL)	Zero-latency, high-concurrency local relational storage for pages, leads, media records, and audit logs.
Cloud Replication	Supabase Cloud PostgreSQL (PostgREST API & SDK)	Serverless database replication ensuring cross-container persistence on serverless deployments (Vercel).
Media Storage & CDN	Supabase Storage (Public media bucket) + Local File System Mirroring	High-speed global content delivery for guide covers, country banners, and high-resolution marketing media.

1.2 Dual-Tier Database Persistence Engine

To deliver maximum reliability across both dedicated VPS environments and modern serverless platforms (e.g., Vercel), the architecture employs a **Dual-Tier Persistence Engine**:

- SQLite Local Storage (Primary Engine):** Configured with Write-Ahead Logging (`journal_mode = WAL`) and foreign keys enforced. Handles high-speed reads and local writes with zero network latency.

- **Supabase Cloud Synchronization:** When running in serverless container environments where local disk writes are ephemeral, the platform synchronizes posts, categories, media metadata, and session tokens directly with Supabase Cloud PostgreSQL. Both databases maintain identical schemas.

2. Autonomous Content Publishing & Editorial Engine

The platform includes an **Autonomous Niche Publishing Engine** engineered exclusively for Kishaa International's specialized practice areas. The engine automates the entire lifecycle of professional editorial content creation—from policy research to statutory verification, long-form content generation, high-resolution visual synthesis, cloud media upload, and sitemap registration.

2.1 The 4 Core Practice Pillars

**1. Study Abroad**

UK 2027 Intakes (CAS issuance, graduate visas), Canada university admissions & study permits, Germany tuition-free universities, Italy DSU scholarships, Finland family residence permits, and Georgia medical admissions (MBBS/BDS without IELTS).

**2. Immigration & Visas**

Canada 10-Year Multiple Entry Visit Visa (IRCC Invitation Letters of Intent & home ties), UAE Golden Visa (AED 30k salary criteria), Green Visa, and Schengen Business/Tourist 4-pillar refusal prevention files.

**3. Statutory Proof of Funds**

UKVI 28-day rule (£1,483/mo Inner London vs £1,136/mo Outer London), IRCC CAD \$20,635 update, Germany BAföG €11,904 blocked account, family sponsorship affidavits, and legitimate source of wealth documentation.

**4. IELTS & Cambridge Training**

Structured roadmaps for advancing from Band 6.5 to 7.5 in Academic IELTS, Writing Task 2 coherence frameworks, Speaking fluency under pressure, and Medium of Instruction (MOI) university waivers.

2.2 Automated Article Structure & EEAT Standards

- Every publication conforms to strict Experience, Expertise, Authoritativeness, and Trustworthiness (EEAT) criteria:
- Answer-First Summary:** A 40-to-60 word paragraph answering the exact statutory query, naming legal thresholds, effective dates, and official authorities. Formatted to feed Google Answer Engine Optimization (AEO) and Generative Engine Optimization (GEO) snippets.
 - Key Takeaways:** 4 to 6 scannable, bulleted insights displayed directly under the answer block.
 - Regulatory Comparison Tables:** Structured HTML tables mapping cities, monthly maintenance rates, 9-month totals, and provider comparisons.
 - Counseling Commentary by Zeb Khan:** An authoritative commentary quoting Zeb Khan (CEO & Chief Consultant) advising applicants on evidentiary documentation, avoiding unvetted third-party deposits, and mitigating refusal risks.
 - Automated Read Time Calculation:** Derived dynamically on save at a standard pace of 200 words per minute.

2.3 Hands-Free Media Synthesis & Supabase Storage CDN

Manual image prompt engineering and uploading have been removed from the blog creation interface in favor of a **fully automated visual pipeline**:

- The publishing engine derives an editorial visual prompt reflecting the article theme (e.g. 16:9 widescreen photorealistic architectural capture of international students at a UK university courtyard).
- The synthesis service renders an ultra-high-resolution visual asset.

3. The image binary buffer is uploaded directly to the **Supabase Storage** cloud bucket under `media/guides/`.
4. The public CDN URL (e.g., `https://jpwgnepvudbqgmrtfxjs.supabase.co/storage/v1/object/public/media/guides/...`) is registered in the database `media` table.
5. The post record's `cover_media_id` and `cover_image` are linked, instantly displaying the cover across the public website, resource index, and RSS feeds.

3. Real-Time Government Statutory Data & Verification

A primary frustration for visa applicants is outdated financial regulations. The Kishaa International platform features a **Real-Time Statutory Rate Verification Engine** that continuously reconciles statutory proof-of-funds rules against official gazettes and immigration authorities.

3.1 Statutory Rates Verified & Maintained

Destination	Authority	Statutory Fund / Rule (2026)	Legal Basis & Payout Rules
United Kingdom	UK Visas and Immigration (UKVI)	£1,483/mo (London) £1,136/mo (Outer London)	Must sit continuously in a regulated bank for 28 consecutive days before filing. Maximum 9 months calculated.
Canada	Immigration, Refugees and Citizenship Canada (IRCC)	CAD \$20,635 living funds + 1st year tuition	Requires Guaranteed Investment Certificate (GIC) from approved Canadian financial institution.
Germany	Federal Foreign Office & BAföG	€11,904 per annum	Deposited in an approved Sperrkonto (Blocked Account), releasing €992 per month upon arrival.
Finland	Finnish Immigration Service (Migri)	€9,600 per annum	Single statutory figure designed to cover the student together with accompanying dependants.
Australia	Department of Home Affairs	AUD \$29,710 living costs + OSHC	Subclass 500 Genuine Student (GS) standard with proof of unencumbered access to tuition and travel funds.
United Arab Emirates	GDRFA & ICP	AED 30,000 monthly salary or AED 2M real estate	10-Year Golden Visa residency without local national sponsor; 5-Year Green Visa freelance option.

3.2 Real-Time Cost Calculation Engine

Integrated into the public portal is the **Interactive Itemized Cost Planner** (`/cost-planner`). The calculator disaggregates student relocation expenses into three non-fungible components:

- Statutory Bank Statement Requirement:** The non-negotiable legal sum required by the embassy.
- Monthly City Living Costs:** Granular estimates spanning rent, utilities, food, and local transit across budget, standard, and comfortable tiers.
- Dynamic Seasonal Airfare Matrix:** Real-time flight estimates between Pakistani origin airports (Islamabad, Lahore, Karachi) and global destinations, adjusting for seasonal peak multipliers (summer and December surges).

4. Lead Management, CRM & Reputation Pipeline

4.1 12-Field Consultation Pipeline

Consultation requests submitted through public forms (interactive modal, header quick-book, or destination pages) enter a structured processing pipeline:

- **Idempotency Deduplication:** Every client session generates a unique cryptographic idempotency key. Resubmissions or network retries are automatically collapsed, preventing duplicate leads and staff confusion.
- **Audit Timeline Logging:** Every status transition (e.g., *new* → *assigned* → *contacted* → *consultation scheduled* → *enrolled*) writes an immutable entry into the `lead_activities` ledger with timestamp, actor name, and channel notes.
- **Country Office Routing:** Leads are automatically tagged and routed based on applicant location and destination (Islamabad, Lahore, or Dubai operations desks).

4.2 Automated Trustpilot Review Invitation Protocol

To build transparent public trust, the platform integrates directly with the **Trustpilot Invitation API** via business integration key `iJfstNF7fpBZzmAj`.

Automated Trigger Point: Whenever an applicant submits an enquiry or schedules a consultation with a valid email address, the system automatically triggers a verified review invitation.

Technical Integration Points:

1. **Script Registration in Head:** Loaded asynchronously across all public pages in both the server-rendered template (`views/site/layout.html`) and the React SPA application (`client/index.html`) via `https://invitejs.trustpilot.com/tp.min.js`.
2. **React Client Form Hook:** Triggered in `client/src/components/site/EnquiryForm.tsx` upon mutation success, passing recipient name, recipient email, reference order ID, and interest category tags.
3. **Server-Rendered Confirmation Fallback:** Triggered on the standalone `/thank-you` confirmation route (`views/site/thankyou.html`) for standard form submissions.

4.3 Candidate Case & Profile Examiner Suite

Located within the administrative workspace (`/admin/ai-agent`), the **Candidate Case Examiner** enables counselors to assess prospective students or visa applicants prior to consultation:

- Evaluates academic background, GPA, and graduation year against university admission criteria.
- Analyzes study or employment gap years, recommending appropriate documentation (experience letters, tax filings, or professional affidavits).
- Calculates financial capacity against statutory blocked account and maintenance fund requirements.
- Generates a comprehensive counseling strategy and tailored communication draft with one click.

5. Security, Anti-Indexing & Administrative Governance

5.1 Multi-Layer Googlebot Anti-Indexing Architecture

To guarantee that administrative workspaces, draft revisions, internal APIs, and counselor notes are never indexed by Google, Bing, or web scrapers, the platform implements a **three-tier defense-in-depth protection model**:

Layer	Implementation Mechanism	Technical Effect
Layer 1: HTTP Headers	X-Robots-Tag: noindex, nofollow, noarchive, nosnippet, noimageindex in src/server.js	Enforced at the network socket level for all paths matching /admin*, /api*, and /preview*. Prevents caching, snippets, and indexing even if crawled.
Layer 2: Robots.txt Protocol	Specific user-agent blocks in src/lib/seo.js for *, Googlebot, and Googlebot-Image	Disallows crawling of /admin/, /api/, /lead/, and draft preview paths across all search crawlers.
Layer 3: HTML Meta Tags	Injected <meta name="robots"> and <meta name="googlebot"> in admin layouts	Serves as client-side fallback in views/admin/layout.html and views/admin/login.html.

5.2 Role-Based Access Control (RBAC)

Every administrative endpoint enforces server-side role validation. User permissions are verified cryptographically on the server, not merely toggled in the visual UI:

- Administrator:** Unrestricted system control, settings management, user provisioning, database backups.
- Content Editor:** Guide creation, revision management, taxonomy editing, media library management.
- Lead Manager (Dubai / Pakistan):** Region-scoped lead lifecycle management, timeline notes, client communications.
- Rule Reviewer:** Versioned statutory funds management, compliance verification, and regulatory auditing.
- Trainer:** Cambridge & IELTS curriculum oversight and candidate diagnostic assessments.
- Contributor:** Draft submission without publishing privileges.

5.3 Serverless Session Persistence

Traditional Node.js applications lose memory-stored sessions when deployed to serverless environments (e.g., Vercel container recycling). This platform implements **dual-persisted asynchronous sessions**: session tokens are written simultaneously to local SQLite and the cloud Supabase sessions table with explicit expiration dates. Counselors remain authenticated seamlessly across serverless container restarts.

6. The Platform "Extras" & Custom Built-ins

Beyond standard content management and corporate web presentation, the Kishaa International platform incorporates specialized enterprise capabilities custom-built for high-volume immigration consultancy operations:

#	Custom Capability / "Extra"	Technical Implementation & Business Value
1	Automated Supabase Storage CDN Upload	Eliminates manual image uploads. Synthesized editorial visuals are uploaded directly to the cloud storage bucket and served through high-speed edge CDN endpoints.
2	Automated Trustpilot Reputation Trigger	Automatically captures candidate review invitations at the exact moment of enquiry submission, maximizing organic 5-star ratings without manual staff outreach.
3	Autonomous Niche Autoblogger	Hands-free publishing system that researches, structures, generates, illustrates, and publishes daily EEAT guides on a scheduled background cadence.
4	Live Destination Cost Verifier	Queries real-time government and central bank data on demand, providing instant proof-of-funds rules and living expense breakdowns.
5	Dual-Tier Database Synchronization	Zero-latency local SQLite operations combined with automatic cloud PostgreSQL replication for serverless scaling.
6	Candidate Profile Examiner	Instant evaluation of applicant GPA, gap years, and budget against current immigration laws, providing a pre-consultation counseling roadmap.
7	One-Click Guide & Cover Generator	Blog editor feature that takes a single topic keyword and automatically generates title, SEO tags, scannable summary, body with tables, and cover thumbnail in under 15 seconds.
8	High-Intent Lead Notifications	Immediate email and messaging dispatch whenever an applicant with high-value funding capacity submits an enquiry.
9	Static Site Search with Full Reindexing	Integrated search index updated automatically on every publication, enabling sub-millisecond client queries across guides and countries.
10	Multi-Layer Anti-Indexing Defense	Guarantees complete invisibility of CMS, counselor notes, and private leads from Googlebot and competitive scrapers.

6.1 Operational Testing & Verification Suite

The platform maintains a comprehensive automated test suite (`npm test`) verifying 14 critical enterprise acceptance criteria (`A01–A14`):

- **A01:** Complete public markup and schema.org validation.
- **A02:** Server-side enquiry validation and cross-site request forgery (CSRF) rejection.
- **A03:** Lead creation, activity timeline entry, and staff notification dispatch.
- **A04:** Idempotency deduplication for duplicate submissions.
- **A05:** Sourced cost planner calculations with separated statutory fund rules.

- **A06:** Versioned statutory funds rule approval lifecycle.
- **A07:** Social feed and testimonial moderation status.
- **A08:** Canonical tags, XML sitemap generation, and robots protocol.
- **A09:** Private route exclusion from indexing and one-hop redirect rules.
- **A10:** CMS authentication, cryptographic sessions, and CSRF protection.
- **A11:** Server-enforced role and permission boundaries.
- **A12:** Revision history, autosave working copies, and draft previews.
- **A13:** Search index reflecting published content only.
- **A14:** Comprehensive audit trail, operational tooling, and database backup verification.